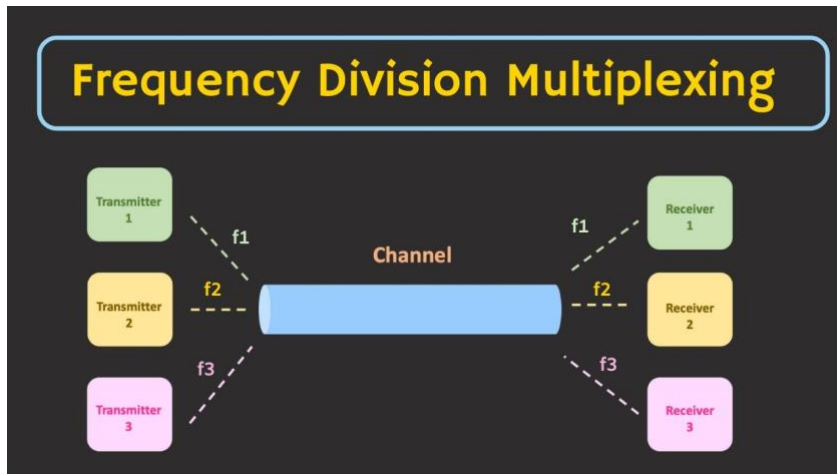


MCS-231 : MOBILE COMPUTING

What is Frequency Division Multiplexing (FDM) ?

Frequency division multiplexing (FDM) is a method for carrying multiple channels of information on one channel by dividing the available bandwidth among the channels. frequency-division multiplexing (FDM), **multiple signals are combined for transmission on a single communications line or channel, with each signal assigned to a different frequency (subchannel) within the main channel.**



Advantages:

- It does not need synchronization between its transmitter and receiver.
- Frequency division multiplexing (FDM) is simpler and easy demodulation.
- Due to slow narrow band fading only one channel gets affected.
- It is used for analog signals.
- A large number of signals (channels) can be transmitted simultaneously.

Disadvantages:

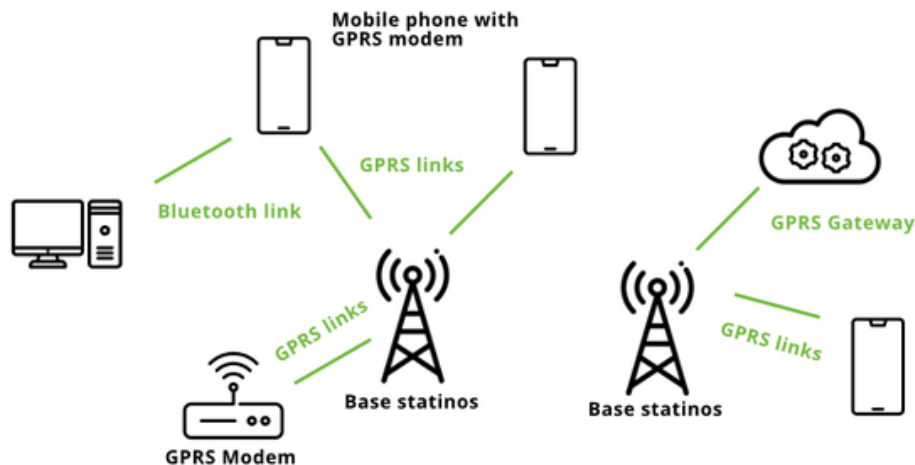
- It suffers problem of cross-talk.
- It is used only when a few low speed channels are desired.
- Intermodulation distortion takes place.

Applications:

- It is used to public telephones and in cable TV systems.
- It is used in broad casting.
- It is used in AM and FM broadcasting.

What is GPRS ?

GPRS stands for general packet radio service. This technology allows information to be sent back and also for forth across a mobile network. It is an addition to today's circuit-switched the data and also provide short message services. **General Packet Radio Service** is a packet oriented mobile data standard on the 2G and 3G cellular communication network's global system



for mobile communications. GPRS was established by European Telecommunications Standards Institute in response to the earlier CDPD and i-mode packet-switched cellular technologies. **What is a GPRS used for?** General Packet Radio Services (GPRS) is a best-effort packet-switching protocol for wireless and cellular network communication services. It is considered best effort because all packets are given the same priority and the delivery of packets isn't

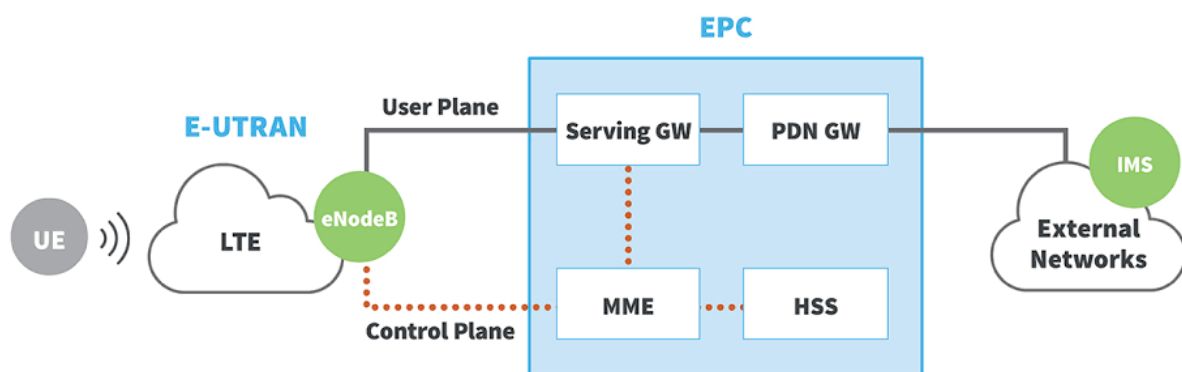
guaranteed. **Features:-** **1.** The maximum theoretical speed can be attained until 171.2kbps by utilizing all eight-time slots simultaneously. **2.** Instantaneous services can be offered by GPRS, and information can be exchanged immediately as per the requirement and relied on coverage and strength. **3.** The application development for GPRS can be allowed for mobile phone range from web browser access, surfing on cyberspace, chat, transfer of attachments. **4.** It also enables mobile application created to achieve faster and better application which is not available in prior

generations. **Services of gprs:-** GPRS extends the GSM Packet circuit switched data capabilities and makes the following services possible:

- SMS messaging and broadcasting.
- "Always on" internet access.
- Multimedia messaging service (MMS)
- Push-to-talk over cellular (PoC) • Instant messaging and presence—wireless village.

What is the LTE?

LTE stands for Long Term Evolution and is sometimes referred to as 4G LTE. It's a standard for wireless data transmission that allows you to download your favorite music, websites, and video really fast—much faster than you could with the previous technology, 3G. Play video. **LTE is a technology for wireless broadband communication for mobile devices and is used by phone carriers to deliver wireless data to a consumer's phone.** Over the previous iteration of 3G, LTE provided high speed, higher efficiency, peak data rates and flexibility in bandwidth and frequency. LTE (Long-Term Evolution) is a fourth-generation (4G) wireless standard that provides increased network capacity and speed for cellphones and other cellular devices compared with third-generation (3G) technology.

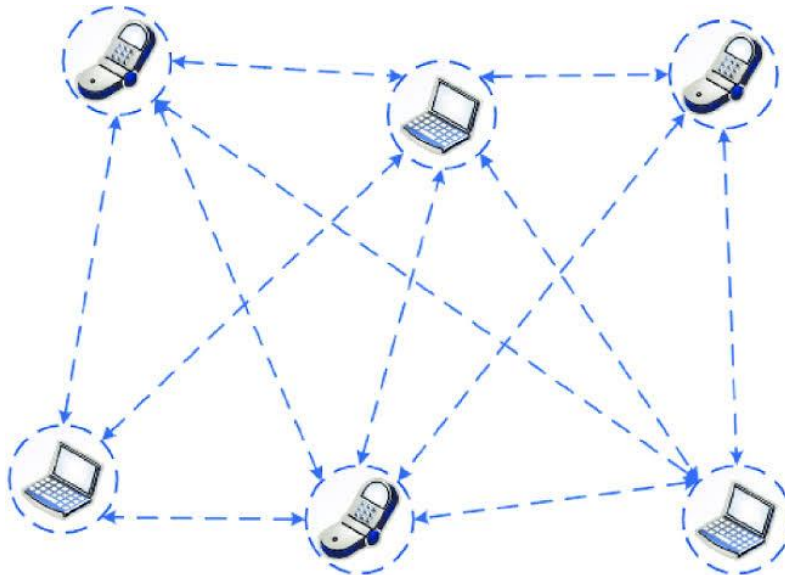


Features:-

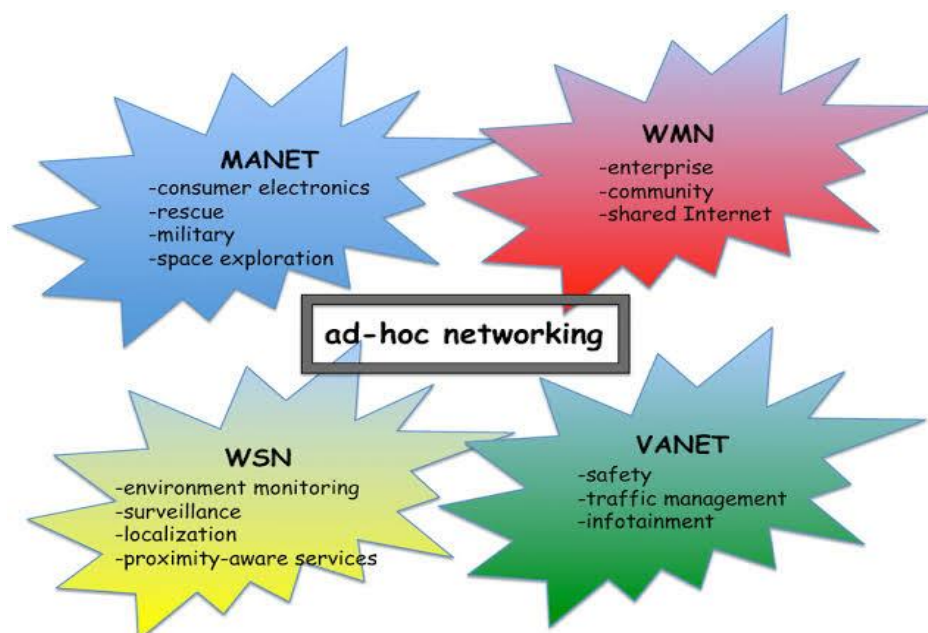
- Better download speed.
- Extremely high voice quality.
- Easy access to Internet, IM, social networks, streaming media, video calling.
- Higher bandwidth.
- Much faster than 3G.

Adhoc Networks:-

A wireless network that allows easy connection establishment between wireless client devices in the same physical area without the use of an infrastructure device, such as an access point or a base station. A typical example of an ad-hoc network is connecting two or more laptops (or other supported devices) to each other directly without any central access point, either wirelessly or using a cable. When to use an ad-hoc network: If you want to quickly set up a peer-to-peer (P2P) network between two devices.

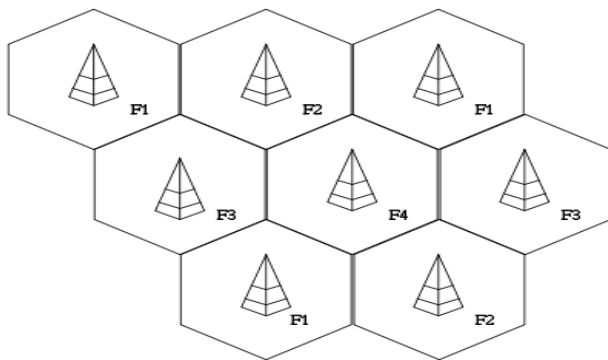


The **advantages** of an ad hoc network include: • Separation from central network administration. Self-configuring nodes are also routers. Self-healing through continuous re-configuration. Scalability incorporates the addition of more nodes. **applications of adhoc network?** Commercial Use: Local Conferences, Events, Seminars. Industrial sector. Emergency services like earthquakes, disaster relief, fire fighting, natural disasters which need services without infrastructure. Education Sector: In classrooms, labs of schools, colleges, etc. **Types of adhoc network:-**



What is a cellular network ?

Cellular network is most commonly the native connectivity method deployed with most cell phones, smartphones, and dial-up devices. It's a mobile-based network used with a radio antenna. These devices are associated with "cell"s and can move from cell to cell in order to communicate and maintain communication. A cellular network or mobile network is a telecommunications network where the link to and from end nodes is wireless and the network is distributed over land areas called cells, each served by at least one fixed-location transceiver. Popular **examples** for cellular networks are GSM (Global System for Mobile communication), GPRS (General Packet Radio Service), CDMA (Code Division Multiple Access) and 3GSM.

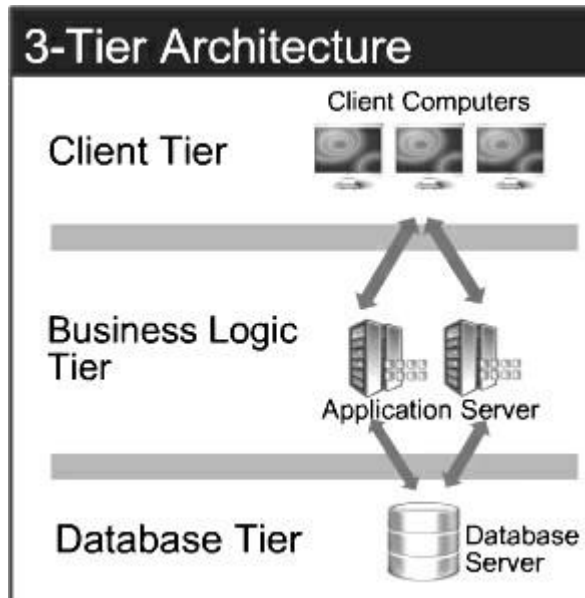


A cellular communication system consists of four major components—namely, a public switched telephone network (PSTN), a mobile telephone switching office (MTSO), The central switch that controls the entire operation of a cellular system., and mobile subscriber units (MSU).

1. **PSTN:-** The public switched telephone network is the aggregate of the world's telephone networks that are operated by national, regional, or local telephony operators. It provides infrastructure and services for public telecommunication.
2. **MTSO:-** The central switch that controls the entire operation of a cellular system.
3. A Cell Site refers to the antenna and ground equipment used to transmit cell phone signals to and from the mobile phone back to the receiver. This equipment is likely to include transmitters/receivers, GPS, backup power sources, base transceiver station (BTS), and more.
4. **MSU:-** A device capable of accessing a cellular network is referred to as a Mobile System (MS) or a Mobile Subscriber Unit (MSU). Cellular service can be accessed by a multitude of devices including two-way email pagers, personal digital assistants and Palm computers as well as voice-only devices.

What is mobile computing architecture?

Mobile computing architecture refers to the definition of multiple layers between user application interfaces, devices, and network hardware. A well-defined architecture is necessary for systematic calculations and access to data and software objects. **Mobile computing** usually implies wireless transmission but, wireless transmission does not imply mobile computing.



1. **Layer-1 : Presentation Layer (UI)**
2. **Layer-2: Application Layer(AL)**
3. **Layer-3: Data Access Layer(DA)**

Presentation Layer (UI): 1. This layer presents data to the user and optionally permits data manipulation and data entry, also this layer requests the data from Business layer. 2. This layer accomplished through use of Dynamic HTML and client-side data sources and data cursors.

Application Layer (AL): The business logic acts as the server for client requests from workstations. It acts according to Business rules, fetch or insert data through the Data Layer. In turn, it determines what data is needed (and where it is located) and acts as a client in relation to a third tier of programming that might be located on a local or mainframe computer.

Data Access Layer (DA):

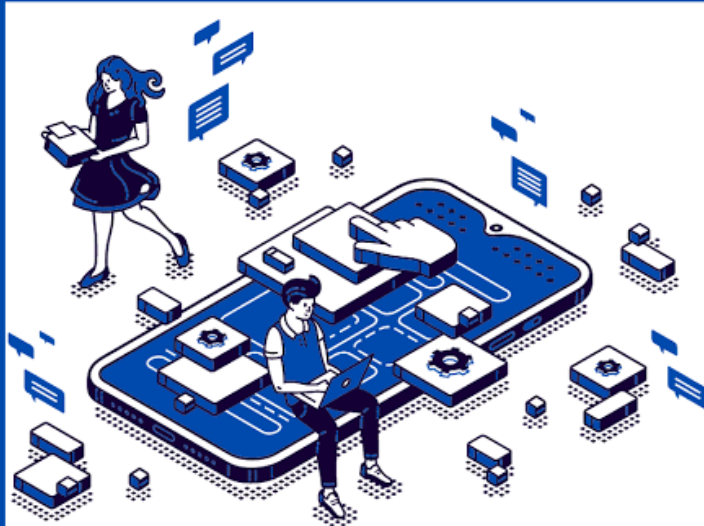
1. The third tier of the 3-tier system is made up of the DBMS that provides all the data for the above two layers.
2. This is the actual DBMS access layer.
3. Avoiding dependencies on the storage mechanisms allows for updates or changes without the application tier clients being affected by or even aware of the change.

Applications of mobile computing:-

Following is a list of some significant fields in which mobile computing is generally applied:

- Web or Internet access.
- Global Position System (GPS).
- Emergency services.
- Entertainment services.
- Educational services.

Applications of Mobile Computing



Exploring Innovative Applications

Social Media Applications

Business Tools

Educational Applications

Gaming Applications

Shopping Applications

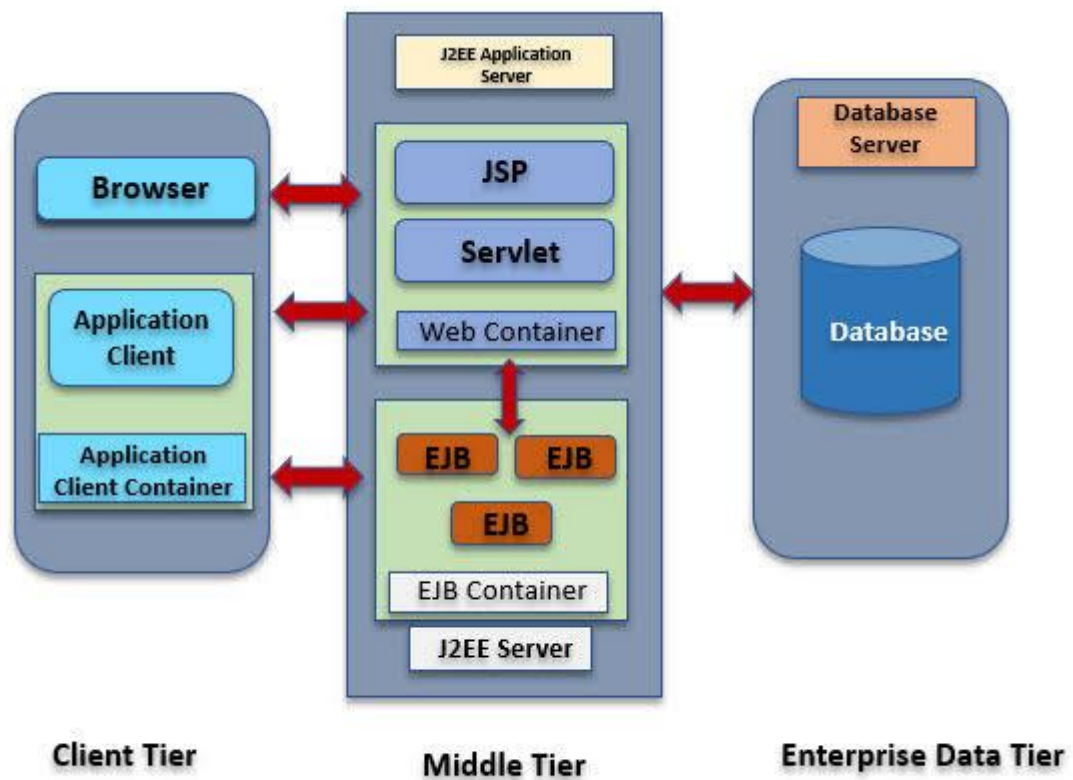
Mobile Security

Explain J2EE architecture.

J2EE stands for Java 2 Platform, Enterprise Edition. J2EE is the standard platform for developing applications in the enterprise and is designed for enterprise applications that run on servers. The JEE platform provides the environment to develop enterprise applications / services using multitier architecture.

- The highly intensified technology made the need for scalable, efficient, faster solutions for information management.
- The JEE technology is rightly apt for meeting these requirements.
- JEE provides a programming development environment which improves the productivity of development, standards for hosting / deploying enterprise applications.

The following are the tiers in JEE application:-



Mobile Operating System:- A mobile operating system allows the user to run other different application software on the mobile, tablets, etc. Moreover, we can say that it is a type of operating system which is specially designed for mobiles, tablets, smartwatches, etc. Furthermore, they are a mixture of computer OS with some additional features for mobiles. Also, they are comparatively light and simple. An operating system (OS) is a program that acts as an interface between the system hardware and the user. Moreover, it handles all the interactions between the software and the hardware. Before knowing different mobile OS, let us study some functions of an operating system.

Functions of OS:- An operating system performs various tasks. Several functions of OS are:

1. Memory Management- It is the management of the main or primary memory. Furthermore, whatever program is executed, it has to be present in the main [memory](#). Therefore, there can be more than one program present at a time. Hence, it is required to manage the memory. The operating system:

1. Allocates and deallocates the memory.
2. Keeps a record of which part of primary memory is used by whom and how much.
3. Distributes the memory while multiprocessing.

2. Processor Management/Scheduling- When more than one process runs on the system the OS decides how and when a process will use the CPU. Hence, the name is also **CPU Scheduling**. The OS:

- Allocates and deallocates processor to the processes.
- Keeps record of CPU status.

3. Device Management- The processes may require devices for their use. This management is done by the OS. The OS:

- Allocates and deallocates devices to different processes.
- keep records of the devices.
- Decides which process can use which device for how much time.

4. File Management

The files on a system are stored in different directories. The OS:

- keeps records of the status and locations of files.
- Allocates and deallocates resources.

5. Security

The OS keeps the system and programs safe and secure through authentication. A user id and password decide the authenticity of the user.

Explain the features of iOS.

iOS stands for “iPhone Operating System.” It is the operating system software that runs on Apple's iPhone and iPad devices. It is similar to the software that runs on a computer, controlling how the device functions and providing a platform for other apps to run on. iOS is the operating system that runs on many of Apple's mobile devices, including the iPhone and iPod Touch. It is the world's second most popular mobile OS, trailing only Android. It is the basis for three other Apple operating systems: iPadOS, tvOS, and watchOS. It is a part of proprietary software. **Advantages of IOS Operating System:-**

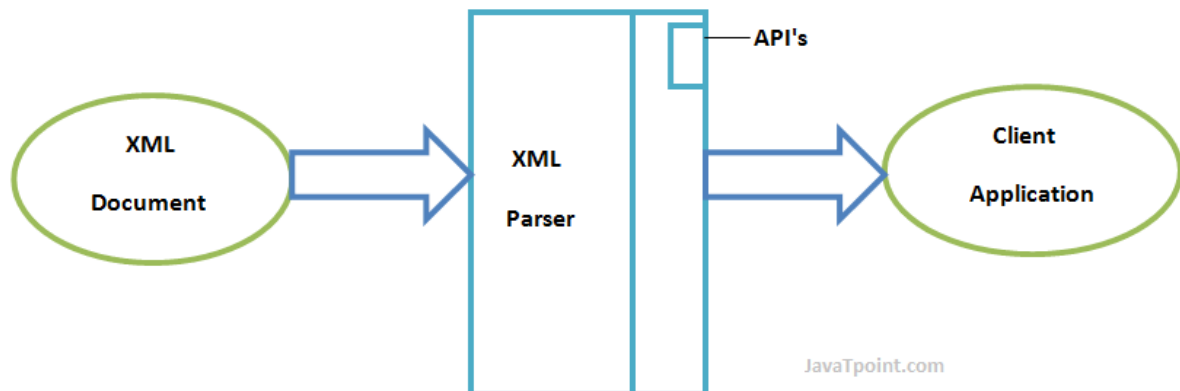
- iOS provides an outstanding user interface and responsiveness.
- iOS is a most secure operating system than any other.
- the iOS operating system is best for Businesses and Professionals.
- Less Heat is Produced as Compared to Android while using.

Features of IOS Platform:-

1. **Multitasking:** iPhone offers multitasking features. It started with iPhone 4, and iPhone 3GS. By using the multitasking feature on an iOS device or using a multi-finger gesture on an iPad, you can swiftly go from one app to another at any moment.
2. **Social Media:** Sharing content and displaying an activity stream are just a few of the ways iOS makes it simple to integrate social network interactions into the app.
3. **iCloud:** Apple's iCloud is a service that offers Internet-based data storage. It works on all Apple devices and has some Windows compatibility, and handles most operations in the background. It is highly encrypted. It offers a backup option to help the user not lose any of their data.
4. **In-App purchase:** In-app purchases, which are available on all Apple platforms, provide users with additional material and services, such as digital items(iOS, iPadOS, macOS, watchOS), subscriptions, and premium content, right within the app. You may even use the App Store to promote and sell in-app purchases.
5. **Game Center:** Game Center, Apple's social gaming network, adds even more pleasure and connection to your games. Game Center provides access to features such as leaderboards, achievements, multiplayer capability, a dashboard, and more.

What is XML?

XML stands for **extensible markup language**. A markup language is a set of codes, or tags, that describes the text in a digital document. The most famous markup language is hypertext markup language (HTML), which is used to format Web pages. XML is a markup language based on Standard Generalized Markup Language (SGML) used for defining markup languages. XML's primary function is to create formats for data that is used to encode information for documentation, database records, transactions and many other types of data.



Features of XML

- Extensible and Human Readable. Even if additional data is added, most XML applications will continue to function as intended.
- Overall Simplicity. ...
- Separates Data from HTML. ...
- Allows XML Validation. ...
- XML Supports Unicode. ...
- Used to Create New Languages.

Advantages

Compatibility

Flexibility

Ease of use

Self-describing

Disadvantages

Complexity

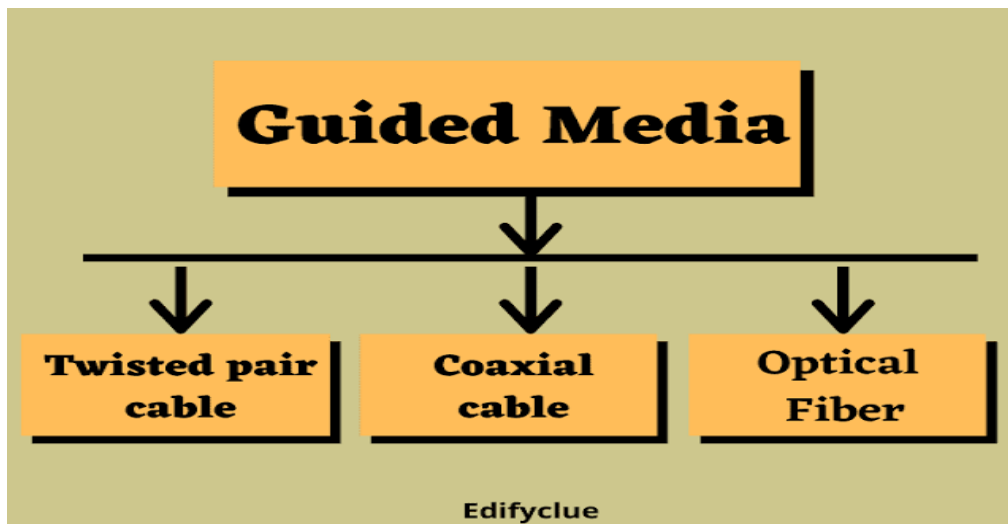
Verbosity

Lack of standardization

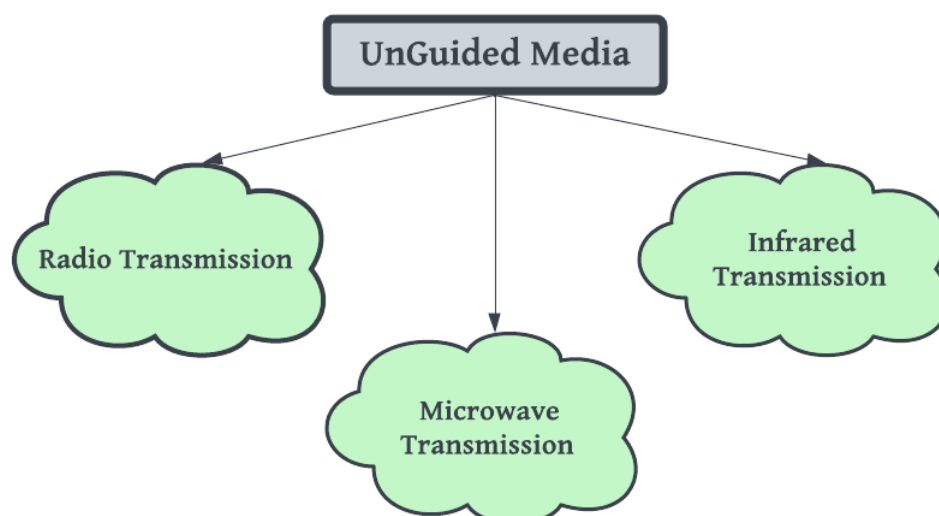
Lack of security

Briefly explain the terms “Guided Transmission” and “Unguided Transmission”.

Guided transmission media consists of physical connection between source and destination through a wire or a cable. There are three basic types of guided media which are as follows – Twisted pair cable. Co-axial cable. Fiber-optic cable. Example - With guided transmission media, the waves are guided along a physical path; examples of guided media include phone lines, twisted pair cables, coaxial cables, and optical fibers.

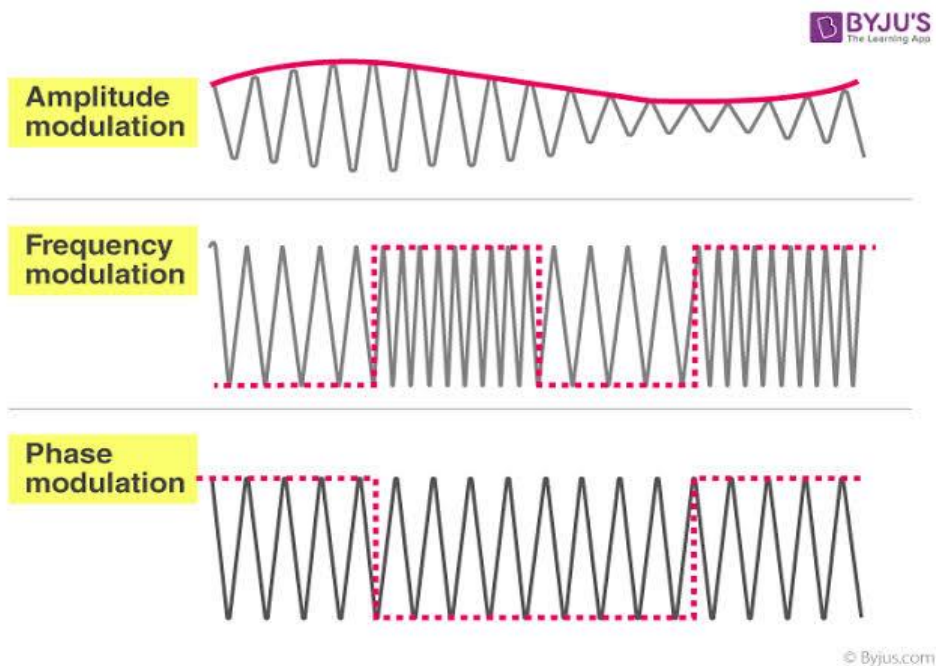


An **unguided transmission** sends electromagnetic waves without the need of a physical medium. As a result, it is sometimes referred to as wireless transmission. Air is the medium through which electromagnetic energy can freely travel in unguided transmission. Unguided transmission media, also called wireless media, are used to transmit data over the air without the use of physical cables. Examples of unguided medium is include microwave, radio or infrared.



What is meant by Modulation?

Modulation is the process of converting data into radio waves by adding information to an electronic or optical carrier signal. A carrier signal is one with a steady waveform -- constant height, or amplitude, and frequency. **For example**, the modulation signal might be an audio signal representing sound from a microphone, a video signal representing moving images from a video camera, or a digital signal



representing a sequence of binary digits, a bitstream from a computer.

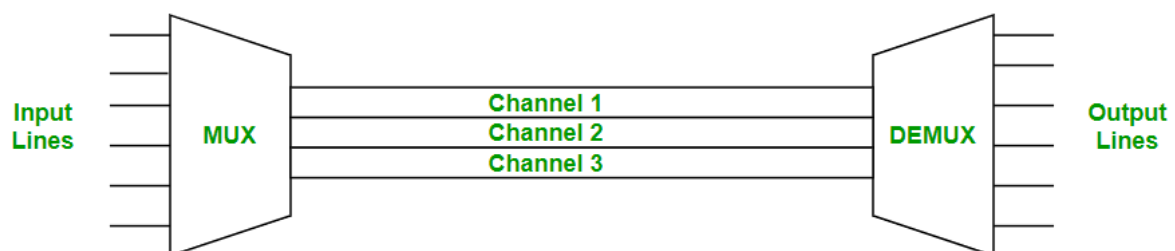
There are three types of modulation:

- Amplitude Modulation.
- Frequency Modulation.
- Phase Modulation.

Amplitude modulation (AM) is a modulation technique used in electronic communication, most commonly for transmitting messages with a radio wave. In amplitude modulation, the amplitude (signal strength) of the wave is varied in proportion to that of the message signal, such as an audio signal. **Frequency Modulation (FM)** is the encoding of information in a carrier wave by changing the instantaneous frequency of the wave. FM technology is widely used in the fields of computing, telecommunications, and signal processing. **Phase modulation (PM)** is a modulation pattern for conditioning communication signals for transmission. It encodes a message signal as variations in the instantaneous phase of a carrier wave. Phase modulation is one of the two principal forms of angle modulation, together with frequency modulation.

What is meant by Multiplexing?

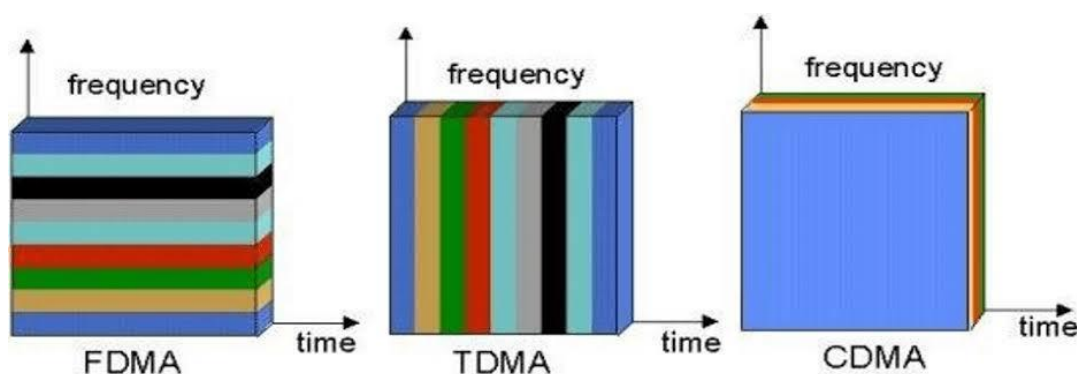
Multiplexing is the technology that is able to combine multiple communication signals together in order for them to traverse an otherwise single signal communication medium simultaneously. Multiplexing can be applied to both analog and digital signals. Multiplexing is the process of combining multiple signals into one signal, over a shared medium. Phone calls are a good example of multiplexing in telecommunications. That is, more than one phone call is transmitted over a single medium. Multiplexing techniques include time-division multiplexing (TDM) and frequency-division multiplexing (FDM).



Time-division multiple access is a channel access method for shared-medium networks. It allows several users to share the same frequency channel by dividing the signal into different time slots. The users transmit in rapid succession, one after the other, each using its own time slot.

Code-division multiple access is a channel access method used by various radio communication technologies. CDMA is an example of multiple access, where several transmitters can send information simultaneously over a single communication channel. This allows several users to share a band of frequencies.

FDMA is another way to enable multiple users to share an entire communication resource. In FDMA (short for frequency division multiple access) each user uses a different band of frequencies to communicate his or her information.



Advantages and disadvantages of 4G network

4G stands for 4th generation, It is long term evolution. Its main objective is that it provides more reliable mobile broadband internet for the device such as a smartphone, tablet, computer, laptops. Speed is too much faster than 3G, so 4G is amazing speed and efficiency and they took the mobile market by storm. So this article gives the advantages and disadvantages of a 4G network to know more about it.

Advantages of a 4G network :

- Better spectral efficiency
- High speed
- High capacity
- High bandwidth
- Tight network security
- High usability: any time, anywhere and any with technology
- Support for multimedia services low transmission cost
- Low cost per bit
- A seamless network of multiple protocols and air interface
- Affordable communication system
- Have easier access to services and application
- Increases the level of use of synchronization
- Machine to machine communication provided
- Global access, service portability, and a variety of quality of services provided

Disadvantages of a 4G network :

- The battery uses is more
- Hard to implement
- Need complected hardware
- It needed to avail services of 4G technology
- The equipment required for a next-generation network is still very expensive
- The network has more problem has security issues
- Not many areas of 4G services yet
- Network protocol and standardization have not to be defined
- High data prize for consumers
- Need different handsets
- Power consumption is high
- Roaming and data or voice work together has not yet been implemented
- Require closer base station and are expensive

